



Syllabus Application



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DSA 301 Data Visualization

Faculty Faculty of Engineering and Natural Sciences

Semester Spring 2025-2026

Course DSA 301 - Data Visualization

Time/Place	<u>Time</u>	<u>Week Day</u>	<u>Place</u>	<u>Date</u>
	08:40-09:30	Mon	FENS-L045	Feb 16-May 22, 2026
	10:40-12:30	Thu	FENS-G035	Feb 16-May 22, 2026

Level of course Undergraduate

Course Credits SU Credit:3, ECTS:6, Engineering:3

Prerequisites DSA 210 or CS 210

Corequisites DSA 301R

Course Type Lecture

Program Requirements >

Instructor(s) Information

Selim Saffet Balcısoy

Email: balcisoy@sabanciuniv.edu

Course Information

Catalog Course Description

In this Data Visualization course, students will embark on an exploration of how to effectively represent and communicate data through visual means. The course begins with foundational concepts, including principles of visual perception and design, to understand how to create intuitive and engaging visualizations. Students will learn to leverage various types of visual representations such as charts, graphs, and interactive dashboards, and will delve into advanced techniques like data mapping, multidimensional visualizations, and real-time data interaction. The course will also cover essential tools and technologies used in the field, including popular visualization libraries and software. Emphasis will be placed on both the theoretical aspects of visualization and practical skills, ensuring students can translate complex data sets into clear, actionable insights. Through hands-on projects and real-world case studies, students will gain experience in designing and implementing effective visualizations that meet the needs of diverse audiences and stakeholders.

Course Learning Outcomes:

Course Objective

By the end of the course, students will be able to:

- Understand Visualization Principles.
- Create and Interpret Visualizations.
- Utilize Data Visualizations.
- Design Interactive Dashboards.
- Analyze and Communicate Data Insights.
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Course Materials

Resources:

Weekly reading material will be shared

Technology Requirements:

Laptop computer and Github account

Assessment

The final grade for this course will be based on:

	Percentage (%)	Number of assessment methods
Final	35%	
Midterm	25%	1
Individual Project	40%	4

Policies

Course Policies:

Plagiarism and academic integrity are very important. Students must work individually unless explicitly allowed. Use of AI/LLMs is only permitted when specified by the instructor and must be properly cited.

Attendance will be monitored and graded. The attendance grading will be on top of regular course grading by increasing or decreasing letter grades.

AI Policies

[Guidelines on the "Use of Generative AI" for students](#)

SU Academic Integrity Statement:

<https://www.sabanciuniv.edu/en/academic-integrity-statement>

Health Report Requirements

[Student Medical Reports Instruction Letter](#) 

Weekly Content

Week	Topic
Week 1	Introduction to Data Visualization
Week 2	Visualization Design, Sketching
Week 3	Visualization Design, Sketching
Week 4	Sensing and Perception
Week 5	Sensing and Perception

Week	Topic
Week 6	Data and Dashboards
Week 7	Visual Prototyping
Week 8	Interactive Grammar of Visualizations: Vega Lite
Week 9	Vega-Altair and other tools
Week 10	Interactive Visualizations
Week 11	Maps
Week 12	Visual Design and Aesthetic Perspective
Week 13	Deceptive Visualizations and Storytelling
Week 14	Storytelling